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Vanadium Contamination of FCC catalyst: Understanding the Destruction and Passivation Mechanisms

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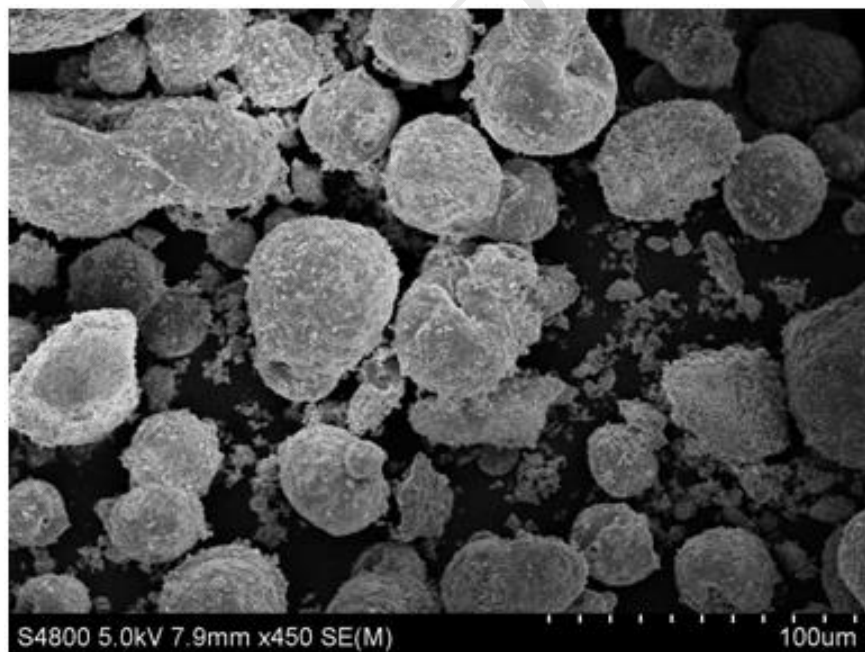
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Graphical Abstract



Research Highlights

- A mixed metal oxide is studied as vanadium passivator for the FCC process
- Vanadium deactivates FCC catalysts by both intra and inter particles migration mechanisms
- Passivation of vanadium proceeds through acid-base chemistry principle by formation of vanadate of the passivator stable to vanadium attack under FCC unit conditions
- A definite amount of passivator material is necessary for maximum performance
- Incorporation of passivator before the metallation provides the best practice

Abstract

Vanadium scavenging material comprising magnesium and yttrium oxides (Mg-Y-oxide) was prepared and evaluated for vanadium capture under simulated FCC regeneration conditions for typical FCC process. Vanadium trapping ability was measured using BET, XRD, FTIR, EDX coupled SEM and MAT. The results show that physiochemical properties, vanadium trapping and catalyst protection increase with amount of the trap material. SEM-EDX results confirm migration of vanadium as indicated by its presence in the passivator particle. Vanadium migrates within the catalyst particles by both inter and intra particles mechanisms. The XRD data further reveal high vanadium selectivity towards yttrium when passivator is incorporated after impregnation of vanadium, with the formation of crystalline yttrium vanadate (YVO_4). Catalytic evaluation indicates improved activities of catalysts containing passivator which correlate with physiochemical parameters. In the passivation mechanism, mobile vanadic acid preferentially reacts more with Y_2O_3 in mixed metal oxide due to its higher activity (active sites), forming YVO_4 . These results therefore reveal that the passivator protects the catalyst by formation of immobile

vanadate with mobile vanadic acid, thereby reducing hydrolysis of the catalyst framework and thus preserving important properties of the catalyst.

Keywords: FCC catalyst, Vanadium, Deactivation, Passivation, Mechanism

1.0 Introduction

Fluid catalytic cracking (FCC) is an important process in the petroleum refinery. It is responsible for 35-45 % gasoline pool in a typical refinery [1-3], and represents the heart of the refinery for production of light fuels [4]. The FCC process involves contacting heavy oil fractions such as vacuum gas oil at high temperature in the presence of a catalyst [5]. This high temperature reaction results in breaking down of large hydrocarbon molecules to lighter fraction of gasoline and diesel, as well as gaseous fuel (liquefied petroleum gas, LPG) and other petrochemical products [6, 7]. Since FCC is a versatile process and can process several feedstock types, including the abundant feeds that are highly contaminated with organic metals such as vanadium and nickel, it has therefore gained prominence in the petroleum industry. The effects of these metals that are present in the FCC feedstock on the activity and selectivity of the cracking catalyst have been widely studied [8-11]. Vanadium is detrimental to catalyst stability and structure, such as the crystallite, whereas nickel promotes dehydrogenation reactions and is selective to coke and lighter gases in lieu of high demand fuel. During regeneration process when coked catalyst is burned off in the regenerator with hot air, vanadium species deposited on the catalyst migrate under hydrothermal conditions and destroy the catalyst structure [12]. This presents a problem of growing concern to the refiner. Wombsbercher et al. [8], noted that vanadium deposited on the catalyst oxidizes to

vanadium (V) oxide and reacts with water vapor present in the regenerator to form vanadic acid as shown in Eq. 1.



As a strong acid, it then reacts with the zeolite catalyst, destroying its structure and activity. Whichever route vanadium penetrates the cracking catalyst, it poisons, leading to the death of the catalyst's active life. For the fact that vanadium compounds cannot be completely removed by pre-treating feedstock or may be a very difficult task altogether, a probable method for controlling vanadium negative effects on the cracking catalyst is to passivate vanadium compounds under typical FCC conditions. Passivation may involve incorporating additives into the cracking catalyst or adding separate particles along with the cracking catalyst [12, 13]. The use of alkaline earth metals such as Mg, Ca and Ba or rare earth metals as passivators for vanadium have been reported to retain activity and improve selectivity of catalyst during processing of feeds with high metal contents [14]. However, in spite of the good trapping capabilities of alkaline earth metals, there still exist problems; the formation of silicate reduces activity for the contaminant metal [12]. For example, magnesium oxide reacts with silicate to form crystalline magnesium silicate according to Eq. 2.



The formation of crystalline metal silicate can be controlled by modifying MgO slurry to preserve the bulk MgO [12]. In this work, commercial MgO was modified with yttrium, a rare earth element, and applied to the passivation of vanadium effects on FCC catalyst. An effort is been made to understand/describe the mechanism of passivation of FCC catalyst by the new metal composite used as passivator in this study. The mechanism of passivation is discussed in terms of acidity and interaction of the passivator particles with the catalyst. A correlation between structural parameters

and activity of protected catalysts is also studied. It is believed that these results could provide new parameters for predicting deactivation of contaminated FCC catalyst.

2. Experimental methods

2.1 Materials and sample preparation

A commercial FCC catalyst, containing REY with surface area; 234 m²/g, total pore volume; 0.20 cm³/g, and RE₂O₃; 5.20 wt.% was used in the present study. Other components of the fresh catalyst (%) include: Na₂O - 0.17; Al₂O₃ - 51.61; SiO₂ - 24.23; Ni - 0.0117 and V - 0.0203. Vanadium was introduced as vanadium naphthenate containing 2.0 wt.% of the metal, dissolved in n-hexane. The catalyst sample was impregnated to a 0.75 wt.% V (nominal amount). In detail, for a 10.0 g catalyst, 3.75 g of vanadium naphthenate was dissolved in 2.0 mL n-hexane and impregnated to 9.925 g fresh FCC catalyst, dried in the oven at 120 °C and calcined in air at 500 °C for 6 h. The calcined sample was thermally treated in 100% steam at 760 °C for 5 h. A mixed metal composite used as passivator, containing magnesium and yttrium oxides, was prepared by our new facile method. Briefly, 5.0 g MgO was dispersed in 40 ml distilled water and stirred for 20 min. Yttrium nitrate (Y(NO₃).6H₂O) (4.75 g) dissolved in 10 ml distilled water was added and the mixture allowed stirring for another 30 min at room temperature. After this time, the white suspension formed was transferred to a water bath at 101 °C and allowed to stir until evaporation to dryness. The white solid obtained was oven dried overnight at 110 °C and calcined in air at 600 °C for 5 h. The passivator was made to 60–80 µm particle size distributions, similar to FCC catalyst particles. The vanadium passivated catalyst samples were prepared by doping different dosages of the passivator, (3-10 wt.%) as (x) wt.% Y-Mg-oxide, to a fixed weight of catalyst impregnated to 0.75 wt.% V content and oven dried at 120 °C. Two control samples, passivator incorporated before impregnation and after calcinations at 550 °C were also prepared.

2.2 Characterization

The XRD patterns of the samples were obtained on PAN analytical X'pert PRO, X-ray diffractometer using Cu-K α radiation generated at 40 kV and 40 mA. The 2θ range was scanned from 5 to 60° with slit parameters. Surface areas and pore size distribution (PSD) were measured using a Micromeritics Tristar 3000 surface area and porosity analyzer at liquid nitrogen temperature. All the samples were degassed at 300 °C under vacuum for 4 h before analysis. The specific surface areas were calculated using the Brunauer–Emmett–Teller (BET) model [15]. The total volume was calculated from the amounts of nitrogen adsorbed at $P/P_0 = 0.98$. The micropore volume and the external surface area were determined from the t-plot analysis. The PSD of the samples were calculated using the Barrett–Joyner–Halenda (BJH) method from adsorption branch [16]. The morphologies of vanadium passivated catalysts were measured with an S-4800 scanning electron microscope (SEM, Japan HQ) combined with energy-dispersive X-ray spectrometry (EDX) accelerating at 0.5–30 kV. Prior to the measurements, the samples were coated with a thin layer of gold. The types of acid sites on the surface of the oxides of the passivator were determined by transmission FTIR spectroscopy of pyridine adsorbed oxides on a Nicolet 6700 spectrometer equipped with a MCT liquid nitrogen cooled detector and a KBr beam splitter. The samples after adsorption of pyridine vapour for 24 h and desorption under vacuum at 0.09 MPa and 150 °C were put into a sample cup (dimension 10 mm and 2.3 mm) and inserted into the measurement chamber. The spectra of the samples were recorded by accumulating 64 scans at 4 cm⁻¹ resolution adopting reflectance mode in the mid-infrared range (1400 to 1600 cm⁻¹). Reported spectra were subtracted from the background of respective samples.

2.3 Catalytic reaction

Catalytic cracking evaluation of the vanadium passivated catalysts was carried out on a MAT unit, using gas oil (with the properties as listed in Table 1) as feedstock. The reaction was conducted using catalyst-to-oil ratio of 4.0 g/g and reactor temperature of 500 ± 2 °C. In a typical reaction, 1.055 ± 0.005 g of feedstock was introduced into the reactor through a syringe pump, after purging the catalyst loaded reactor with 30 ml min^{-1} nitrogen stream for 30 min, within 75 s, followed by nitrogen purging with the same flow rate for 10 min. The products; gas was collected in a gas pack, and liquid in a receiver connected to a cooling bath. The gas products were analyzed by Varian 3800 gas chromatograph (GC) equipped with two detectors: a thermal conductivity detector (TCDs) and a flame ionization detector (FID) for quantitative determination of individual gases. Liquid products were separated by simulated distillation, in the following grades: IBP–216 °C (gasoline), 216–343 °C (diesel) and ≥ 343 °C (heavy oil), on Varian 3800 chromatograph (GC). Catalyst activity (MA) was calculated according to the expression [17]:

$$MA, \text{wt. \%} = \frac{w_f - R(\%) \times w_l}{w_f} \times 100, \quad (3)$$

where, w_f is weight of gas oil feed (g), R is fraction of material boiling above 216 °C in the liquid product and w_l is weight of liquid in the receiver (g). Coke deposited on the catalyst was analyzed using elemental analyzer (Elementar Vario EL III).

3.0 Results and discussion

3.1 Structural properties of passivator

An effective trap for vanadium must have similar physical properties as FCC catalyst, including particle size distribution. Amongst other desirable features are large pores, high pore volume ($> 0.1 \text{ cm}^3/\text{g}$) and surface area [8]. In this work, we prepared a passivator, Y-Mg-oxide, with structural properties given in Fig. 1. A high pore volume has been achieved; this reduces plugging effect of reaction with small pore materials, thereby, enhancing reaction with vanadium poison precursor.

The XRD pattern is indexed according to the JCPDSs (MgO: 01-087-0651 and Y₂O₃: 01-083-0927).

3.2 Metal deactivation of individual catalyst particles and influence of treatment temperature

The matter whether vanadium poison is homogeneously distributed within the FCC catalyst components, or it is selectively deposited on a specific catalyst component has been investigated by Ruiz-Martínez et al. [4, 18]. In a related study, Buurmans et al. [19] investigated the catalytic activity of individual catalyst particle in the presence of contaminant metals. Therefore, to further widen knowledge on the dynamics of deactivation FCC catalyst in the presence of metals (e.g. vanadium), we studied the effect of laboratory treatment conditions on individual components of the FCC catalyst. The results presented in Fig. 2 reveals that the characteristic profiles of the zeolite (REY in the catalyst) within $2\theta=5-35^\circ$ are retained after hydrothermal treatment in 100% steam at 760 °C for 5 h in the absence of vanadium. Other components of the catalyst are severely deactivated at calcination temperatures. Particularly, kaolinite (clay) was completely deactivated at calcinations, agreeing with the report of Buurmans et al. [20]. During laboratory simulation of industrial E-cat according to the method of Mitchel [21], test catalysts for metal contamination studies are normally subjected to calcination in air between 500–600 °C, after oven drying. A catalyst sample was impregnated to 0.9 wt.% vanadium and subjected to heating at different temperatures and time. As shown in Fig. 3, the effect of temperature treatment preceding hydrothermal treatment in the laboratory simulation of catalyst deactivation was studied to realize vanadium mobility and to identify the vanadium species responsible for zeolite destruction. According to the figure, this treatment barely caused damage to the zeolite structure in the catalyst samples. Even at 760 °C calcination temperature, which the zeolite structure should have been

severely affected, with the high vanadium content, since the melting point (690 °C) of vanadium (V) oxide is exceeded and vanadium was expected to be mobile, which can easily reach the framework of the zeolite, there is no remarkable degradation of crystallinity as indicated by retainment of the major reflections of Y-type zeolite [14]. The implication of this result is that vanadium oxide (V_2O_5) in which vanadium exists as in the catalyst under this condition is not the only operating species for the destruction of vanadium contaminated catalyst in the FCC unit.

3.3. Structural analysis of passivated catalysts

The crystalline structure of the catalysts was investigated by XRD analysis of the samples. Fig. 4 shows patterns of catalysts with passivator, incorporated after impregnation of vanadium and drying in the oven, calcined in air at 550 °C for 6 h. The patterns reveal virtually sharp reflections which can be exclusively attributed to the zeolite phase. Vanadium passivated (with different dosages of passivator) and hydrothermally treated catalyst samples are compared with a sample impregnated with vanadium but had no passivator and treated accordingly (Fig. 5(I)). Evidently, Fig. 5(I)-a (without passivator) shows severe collapse of the zeolite structure, in accordance with behavior of unprotected catalyst containing vanadium impurities, after being subjected to hydrothermal treatment [8, 19]. Prominent peaks are those near 2θ 26.6 and 45.7°, corresponding to quartz (SiO_2) and alumina (Al_2O_3) phases, respectively, a characteristic of amorphized sample. A careful dissection of all patterns revealed other peaks identified at *ca.* 23.8, 25.0 and 42.9° 2θ , and corresponding to zeolite (533 phase), YVO_4 (JCPDS – 01-076-1649), and bulk MgO (JCPDS-01-087-0651), respectively. Inspection of all the patterns reveals reduced 42.9° 2θ peak for the samples (Fig. 5(I)-b&e), which may be due to the low concentration (3.0 wt.%) of the passivator in the sample (Fig. 5(I)-b) or a stronger interaction between the vanadium species and the passivator particles (bulk MgO) in the sample incorporated before vanadium impregnation (Fig.

5(I)-e). However, since our aim was to demonstrate the mechanism in which vanadium reaches the zeolite particle of the catalyst, where destruction originates, and the passivator particle as well, we chose to incorporate passivator after vanadium impregnation and drying for the experiments. Contrary to the report [8] that vanadium readily combines with magnesium to form magnesium vanadate ($\text{Mg}_2\text{V}_2\text{O}_7$), our XRD results showed no any traces of crystalline vanadate of magnesium, instead showed vanadate of yttrium (YVO_4) which underscores the role of yttrium oxide in the composite in passivation of vanadium in the present study. Fig. 6 expresses clearly the regions where the peaks emerged due to performance of the passivator. When bulk magnesium oxide was incorporated, magnesium vanadate was detected, and again, there is presence of magnesium silicate, which is not a suitable passivator. These phases are completely absent or minimally present in Y-Mg-oxide incorporated sample (Fig. 6). Vanadium passivation of the catalyst at different conditions is compared in Fig. 7. The results show that a higher degree of protection of the catalyst was realized when passivator is incorporated directly to oven dried sample and steam-aged without calcination. However, this route resulted in higher bulk MgO in the resulting formulation. In any case, passivation is shown to have occurred but severity of the conditions seems influencing passivation effectiveness.

3.4. Surface area and pore structure analysis of passivated catalysts

The degradation of textural properties of the vanadium contaminated catalyst without the passivator (0 wt.% Y-Mg-oxide) is quite obvious. The surface areas and pore size distribution of the samples are presented in Figs. 8 and 9. Both BET surface and micropores surface areas correlate well and increase as the amount of additive increases. On the contrary, mesopores surface area which is mostly associated with the catalyst components except zeolite rarely changed. The results support the insignificant impact of deactivation on these catalyst components by vanadium

or are a behavior of particles that are stable to vanadium activity. The nitrogen adsorption–desorption isotherms of the catalysts are shown in Fig. 10. The catalysts exhibit Type IV isotherm, which is a characteristic of mesoporous materials with H-4 hysteresis loops. The hysteresis loops occur in the range of relative pressure from 0.42 to 1.0, indicating the presence of mesopores. Moreover, there is an increased adsorption of nitrogen in the low pressure region, indicating the filling of more micropores in the passivated sample. The presence of bimodal mesoporous structure is illustrated for the steam-aged catalyst and all of the passivated catalysts according to the pore size distribution plot (Fig. 9). The peak centered around (4.5–5.3 nm) is attributed to matrix components, whereas the peak located near 14.7 nm in steam-aged sample is attributed to zeolite component caused by dealumination of framework alumina, leading to mesopores formation. Contaminated samples showed increased pore size, corresponding to structure impairment. The catalyst sample containing 7.5 wt.% additive incorporated before metallation showed excellent stability against steaming and vanadium impurities. This is evidenced by reducing pore size from *ca.* 23 to 14.8 nm, near 14.7 nm of steam aged sample.

The peak area of 2θ between 22.0–24.5° is generally employed to determine the crystallinity of zeolites and/or zeolitic materials relative to a reference standard zeolite sample, and the BET surface area or more precisely micropore surface area is another metric of crystallinity; the greater the peak or surface area, the more crystalline the catalyst. We aimed to correlate these two metrics as shown in Fig. 11. From the figure, a peak area was approximated for 0 wt.% Y-Mg-oxide (without passivator) sample using Eq. 4.

$$y = 1.51x - 15.36 \quad (4)$$

where “y” is the micropores surface area (calculated from T-plot) and “x” is the peak area (calculated from XRD). From the equation, “x” (10.5 [cts*°2Theta]) for 0 wt.% Y-Mg-oxide

sample was computed within error detection limit and further used to estimate degree of protection as reported in Table 2. It can be seen from the table that approximation of passivator's performance correlates well at high dosage, confirming the proposition that increase in the concentration of passivator enhances its performance. Nevertheless, the 7.5 wt.% additive containing sample (added before metallation) gives the highest peak area of 66.4 against 52.0 [cts*°2Theta] for 10.0 wt.% additive containing sample (incorporated after metallation), in accordance with BET results.

3.5 Surface analysis of catalyst and passivator particles

The catalyst/additive mixture was systematically analyzed on an SEM coupled with EDX, and the resulting images are shown in Fig. 12. Fig. 12a refers to a catalyst microsphere showing spots of smaller additive particles (white patches) on its surface. However, some isolated additive particles can be seen as irregular particles within the bulk mixture (Fig.12b). Fig. 12c shows an enlarged image of additive particles which has been confirmed from elemental scanning (Table 3). Point analysis revealed migration of vanadium from the catalyst particles to the passivator particle by a supposedly inter-particle transfer mechanism earlier reported [14, 22]. In Table 3, distribution of elements (EDX analysis) is shown. It can be observed that Al, Si and O have proximate concentrations and an even distribution at all locations (points 1 to 5). However, vanadium as well as additive elements (Mg and Y) were detected only at points that had high concentrations of passivator. It is worthy of note that for the passivator particles in the catalyst, the elemental composition from EDX would differ from that obtain for fresh passivator particle using XRF (results not shown). The reasons may be due to the fact that EDX is a semi-quantitative method that can only give the elemental content of a small domain (spot), and the composition of the element is calculated base on the total elemental composition, including Al, Si and V. To further verify vanadium migration, elemental scanning was also performed on supposed additive particles

and the results (Table 4) indicate vanadium presence, confirming that vanadium migrated from the catalyst particle to the passivator particle, which reduced its absolute effect on the catalyst activity. Reasonable vanadium amounts are recorded from all points; magnesium showed increased surface concentrations at all locations justifying that the particle is that of the passivator and that more yttrium component of the passivator has interacted with vanadium.

3.6 Evaluation of catalyst

A gas oil feedstock was catalytically cracked on a micro activity (MAT) reactor to test the performance of metallated catalysts containing our prepared passivator. The test results (average of two runs) presented in Table 5, demonstrate effects of additive on cracking performance and product distribution. MAT catalyst activity ranges from 54.8 to 80.6 wt.%, with the pre-passivated and metallated (containing *7.5 wt.% Y-Mg-oxide) catalyst giving the highest activity (Fig. 13a). The insert in Fig. 13b shows the activity of the sample at two different temperatures. The performance of the sample containing 10 wt.% additive is illustrated at different catalyst-to-oil ratios (3.0-5.0). As would be expected, conversion increases with an increase in catalyst-to-oil ratio. This might be due to increase in effective active surface and acids sites of the catalyst for reaction with the feed molecules. Table 5 also compares normalized evaluation results of catalysts with different amount of the additive to that without additive. As can be seen, conversion increases as the additive dosage increases and reaches a maximum with 7.5 wt.% additive content at similar catalyst-to-oil ratio. The same trend is also shown for gasoline yield; however, LPG and coke content display different trends. Whereas, the highest gasoline yield was obtained using a catalyst containing 7.5 wt.% additive incorporated before metallation (*7.5 wt.% Y-Mg-oxide), it required a higher catalyst-to-oil ratio for the same gasoline proportion with a 10 wt.% additive catalyst sample. These results imply that an optimum yield of gasoline is achievable with 7.5 wt.% additive

incorporated before metallation and collaborate our BET and crystallinity results. The above data indicate the overall improvement in the activity of vanadium contaminated catalyst containing passivator in our study. The coke yield increased with the weight of the additive and catalyst-to-oil ratio. However, the increase of 6.7% on 7.5 wt.% additive catalyst compared with catalyst without passivator at constant catalyst-to-oil ratio represents insignificant effects of vanadium on coke production and additive to coking. The pictures of coke showing passivator particles after reaction are given in Fig. 14. Obviously, the passivator particles retain its whitish color which means they are not involved in coke formation. In Fig. 14a, coked sample without additive, the tiny white spots are quartz sand particles used as a preheating bed in the reactor.

3.7 Mechanism of passivation

The stability of a vanadium contaminated catalyst at 760 °C calcination temperature and the migration of vanadium from the catalyst particle to the passivator particle are clear indications that catalyst deactivation by vanadium occurs in the presence of a vanadium poison precursor, vanadic acid [8], which is formed in the regenerator during catalyst regeneration in the commercial units as mentioned earlier according to Eq.1. Because vanadic acid is a strong acid, ($P^{K_a} = 0.05$) [23], it shows the tendency of reaction with a base/basic material. As noted above, our passivator comprises two basic oxides (MgO and Y_2O_3). In theory, MgO is a stronger basic oxide than Y_2O_3 ; however, it lacks active solid catalyst acid center (Lewis acid). Tamura et al. [24] revealed that Y_2O_3 is a stronger solid acid catalyst (owing to presence of Lewis acid site) compared to MgO. This fact is also confirmed in our present study (Fig. 15). Mobile vanadic acid preferentially reacts with Y_2O_3 in bulk Mg-Y-oxide due to its higher acidity (active site) forming immobile vanadium compound (yttrium vanadate – YVO_4). Formation of YVO_4 can partially consume vanadic acid, making it less available to react with catalyst and MgO, and consequently less magnesium

vanadate formed, and less attack on the catalyst framework results. The formation of immobile YVO_4 is revealed by XRD (Fig. 6). The overall mechanism appears, therefore, to be controlled by this metal oxide activity according to the reaction.



4.0 Conclusion

The deactivation and passivation mechanisms of vanadium contaminated FCC catalysts have been studied. A mixed metal oxide comprising magnesium and yttrium was demonstrated as an effective vanadium passivator during a FCC process. The passivation mechanism of the oxide is based on the acid-base chemistry principle. The results reveal that the passivator protects the catalyst by formation of immobile vanadate of yttrium with mobile vanadic acid, which reduces hydrolysis of the catalyst framework and thus, protects important properties of the catalyst. Catalyst samples with additive show quantifiable enhanced activity with the amount of additive, which collaborate physicochemical properties. Nevertheless, incorporating passivator before metallation gives enhanced yield of desirable products. These results can help open further studies of metal oxide activities which could be probable as vanadium passivator for FCC catalyst.

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Figure Legend

Fig. 1 Structural properties of passivator.

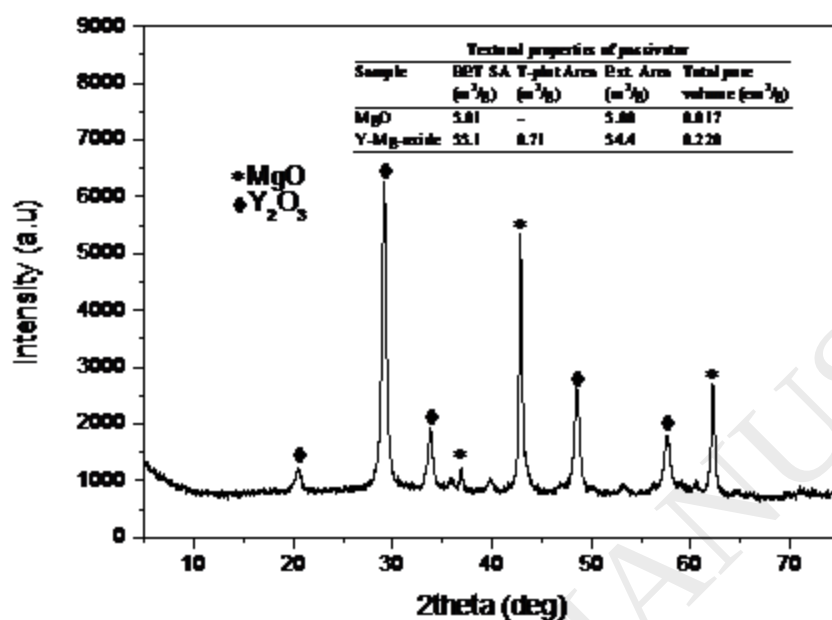


Fig. 2 FCC catalyst components under different treatment conditions.

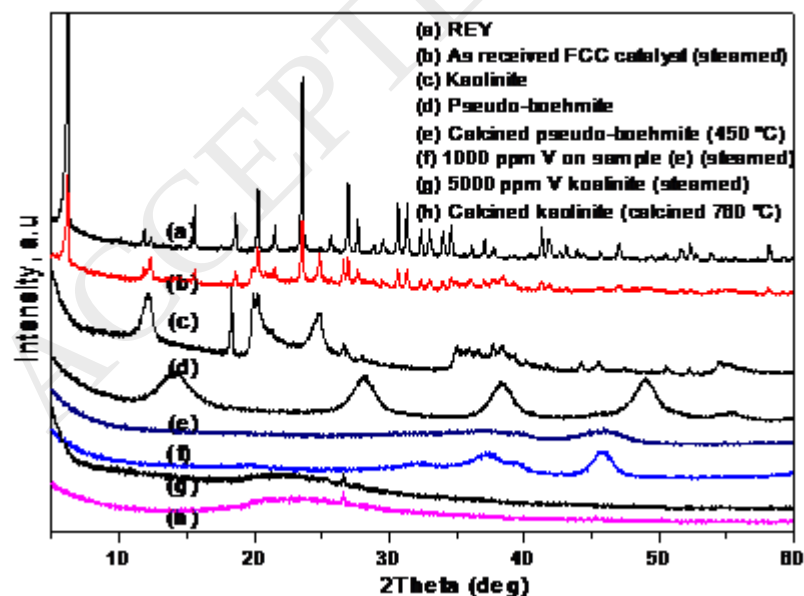


Fig. 3 Effect of calcination temperature on the mobility of vanadium species.

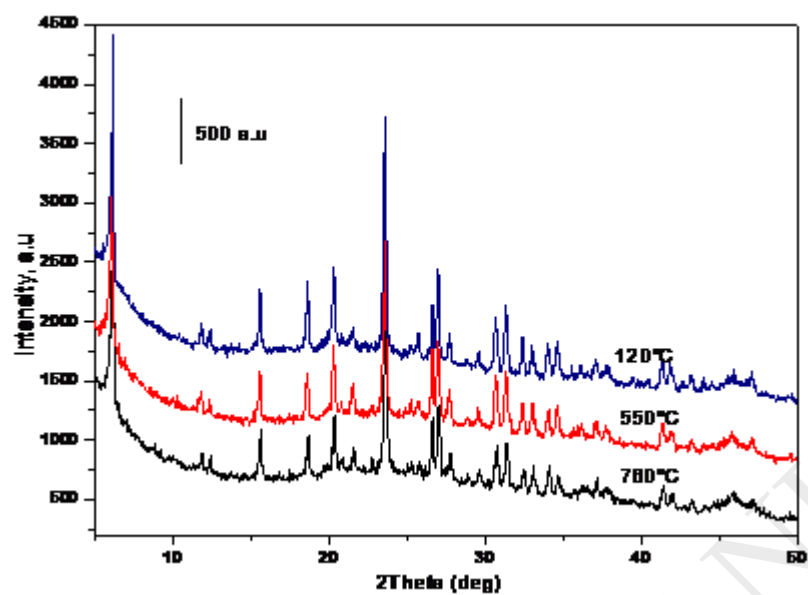


Fig. 4 XRD patterns of calcined (550 °C, 6 h) samples containing passivator. (a–d, f) contain 0, 3, 5, 7.5, 10 wt.% passivator incorporated after drying of vanadium impregnated catalyst at 120 °C for 12 h; (e) 7.5 wt.% passivator incorporated before vanadium impregnation

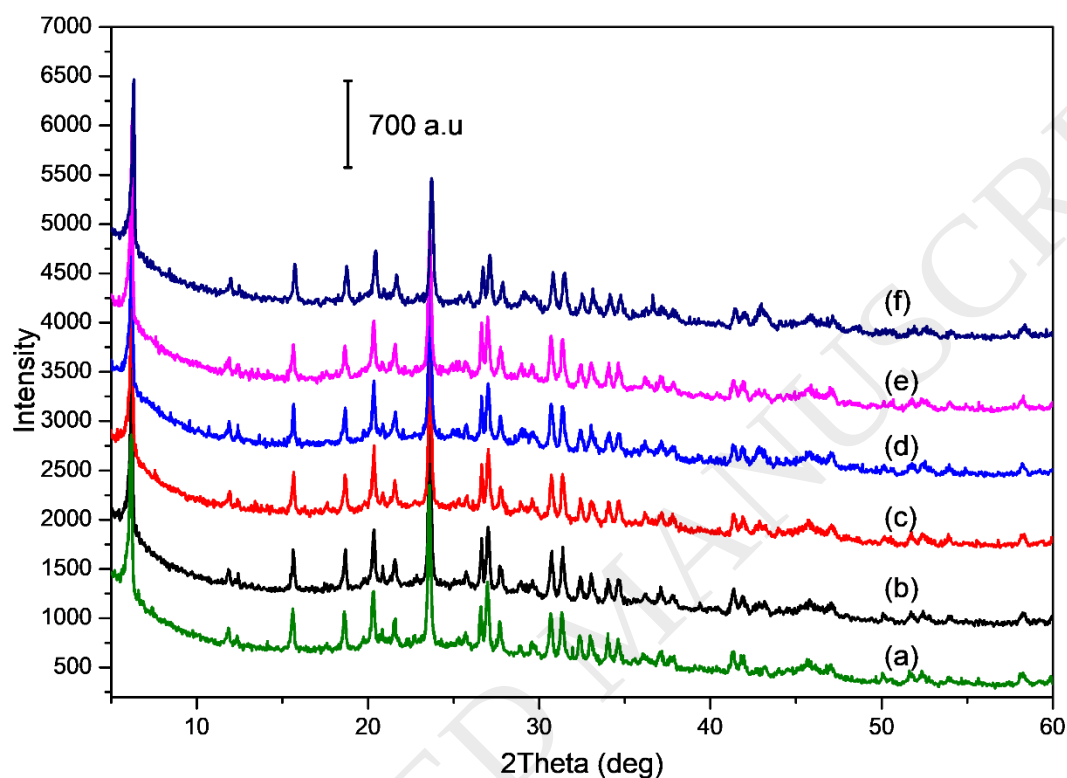


Fig. 5 XRD patterns of: (b–f) calcined (550 °C, 6 h) and hydrothermally treated (ST 760 °C, 5 h) samples with passivator, and (a) sample without passivator treated under similar conditions. Phase alterations/formations due to high temperature and steam treatments and vanadium interaction with passivator.

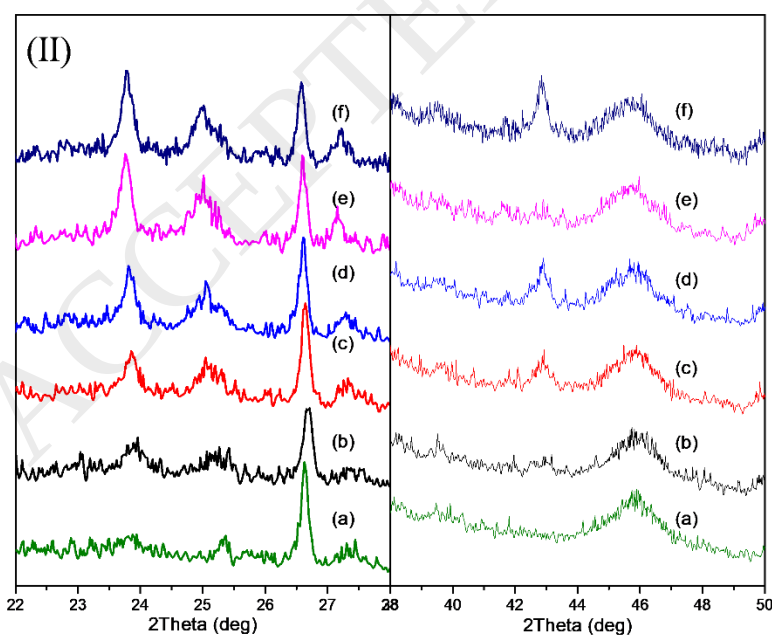
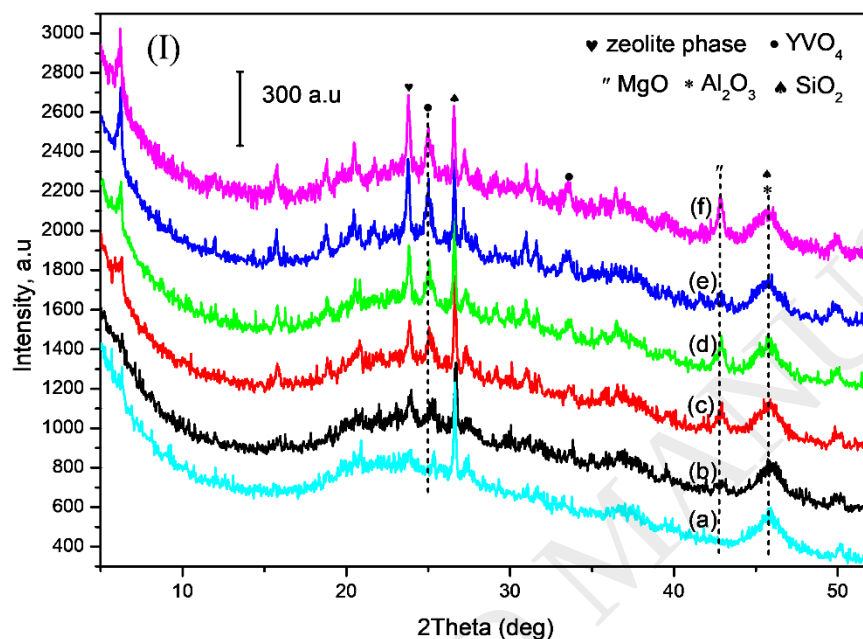


Fig. 6 Phases of compounds formed in the presence of a passivator.

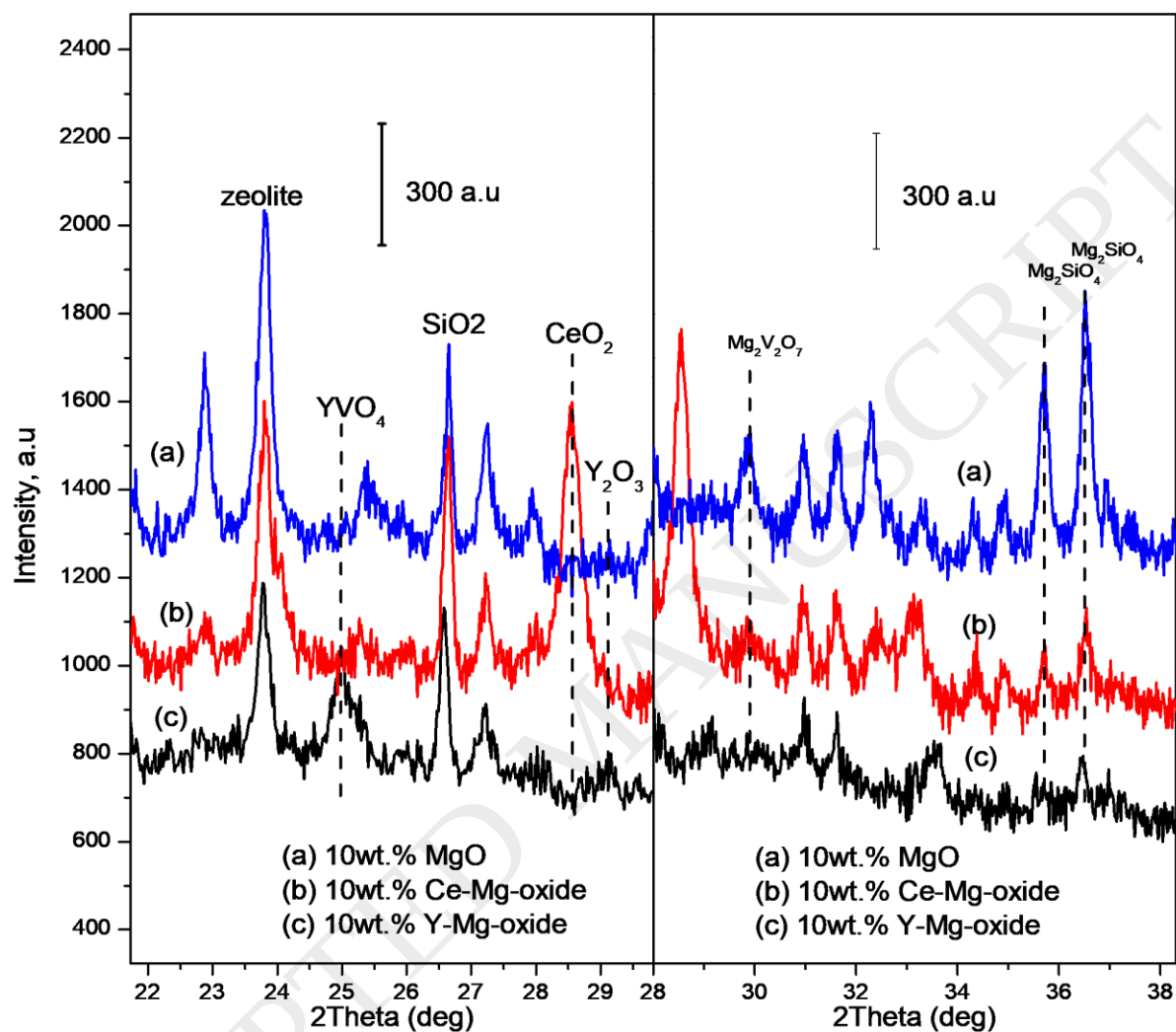


Fig. 7 Comparison of passivation at different treatment conditions. 10 wt.% passivator in catalyst containing 0.75 wt.% V added after (a) calcination 550 °C 6 h (b) drying at 120 °C 24 h and (c) drying at 120 °C 24 h and calcination 550 °C 6 h. All samples were hydrothermally treated (ST 760 °C 5 h).

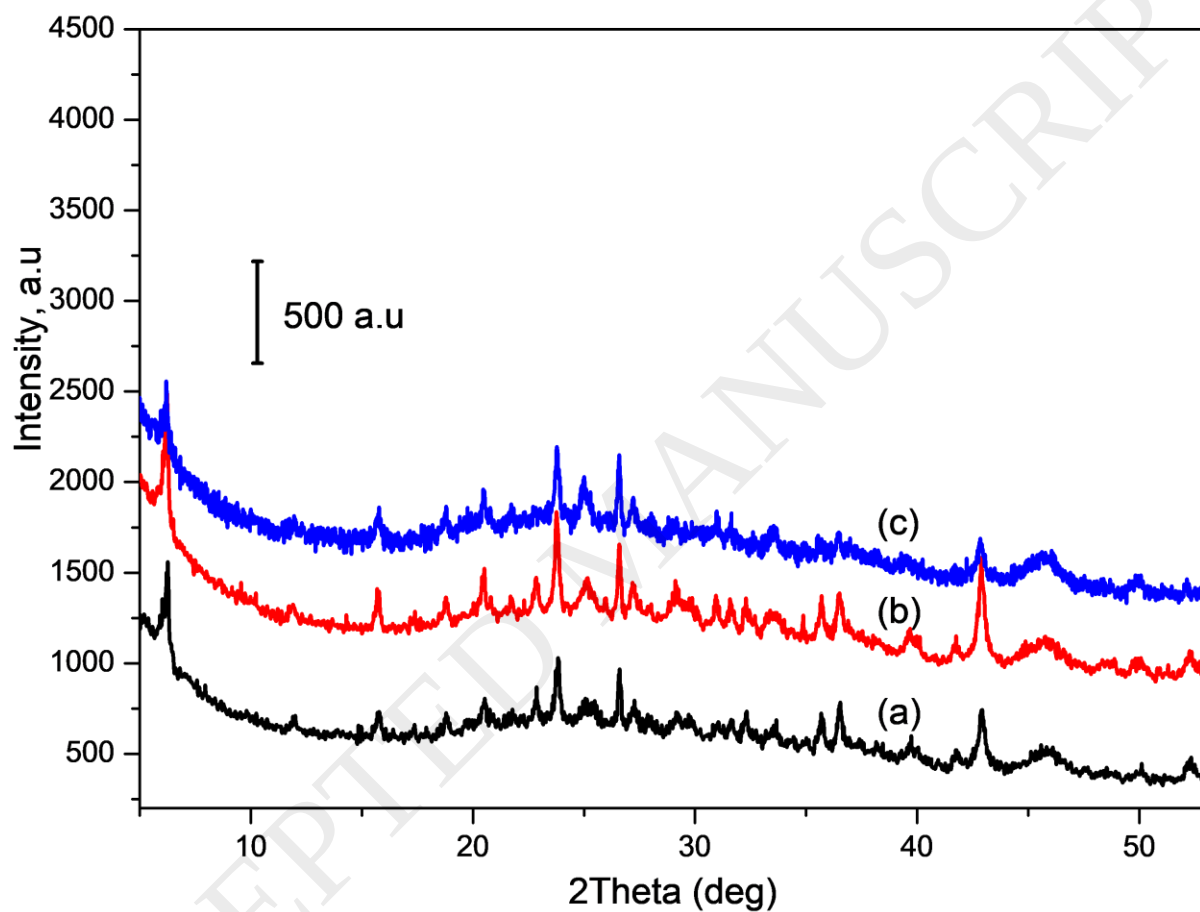


Fig. 8 Effect of passivation on the surface areas of vanadium contaminated catalyst.

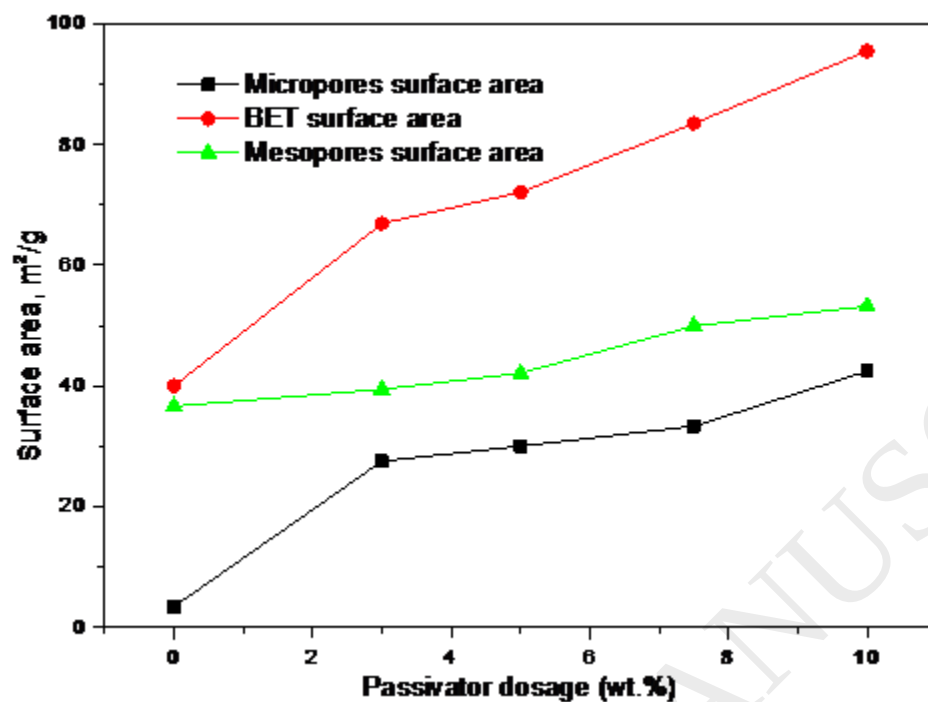


Fig. 9 Pore size distribution calculated from adsorption data.

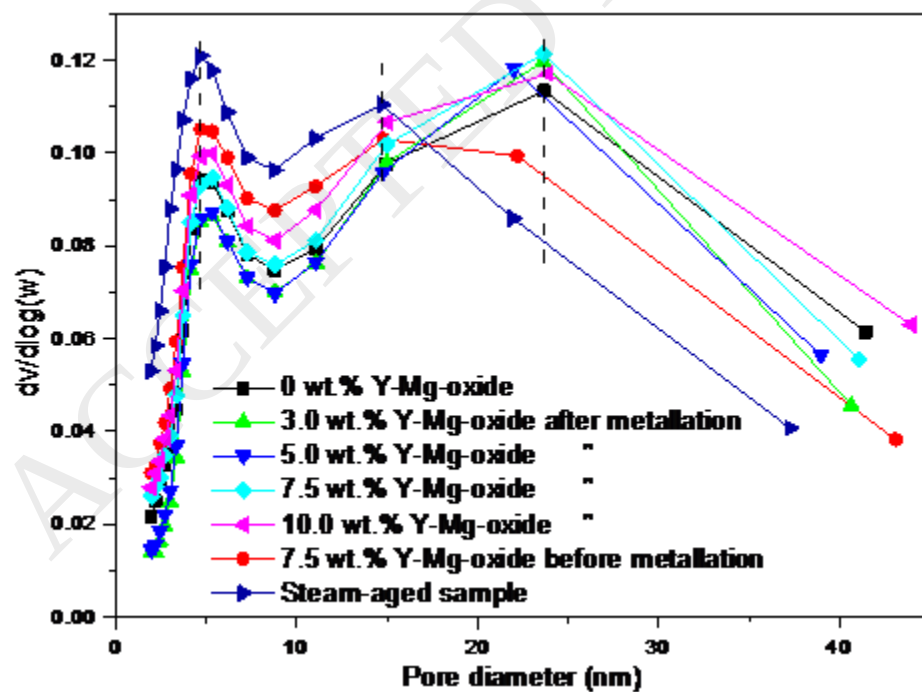


Fig. 10 Nitrogen adsorption-desorption isotherms of catalyst samples: (a) steam-aged (b) 7.5 wt.% Y-Mg-oxide before metallation (c) 0 wt.% Y-Mg-oxide.

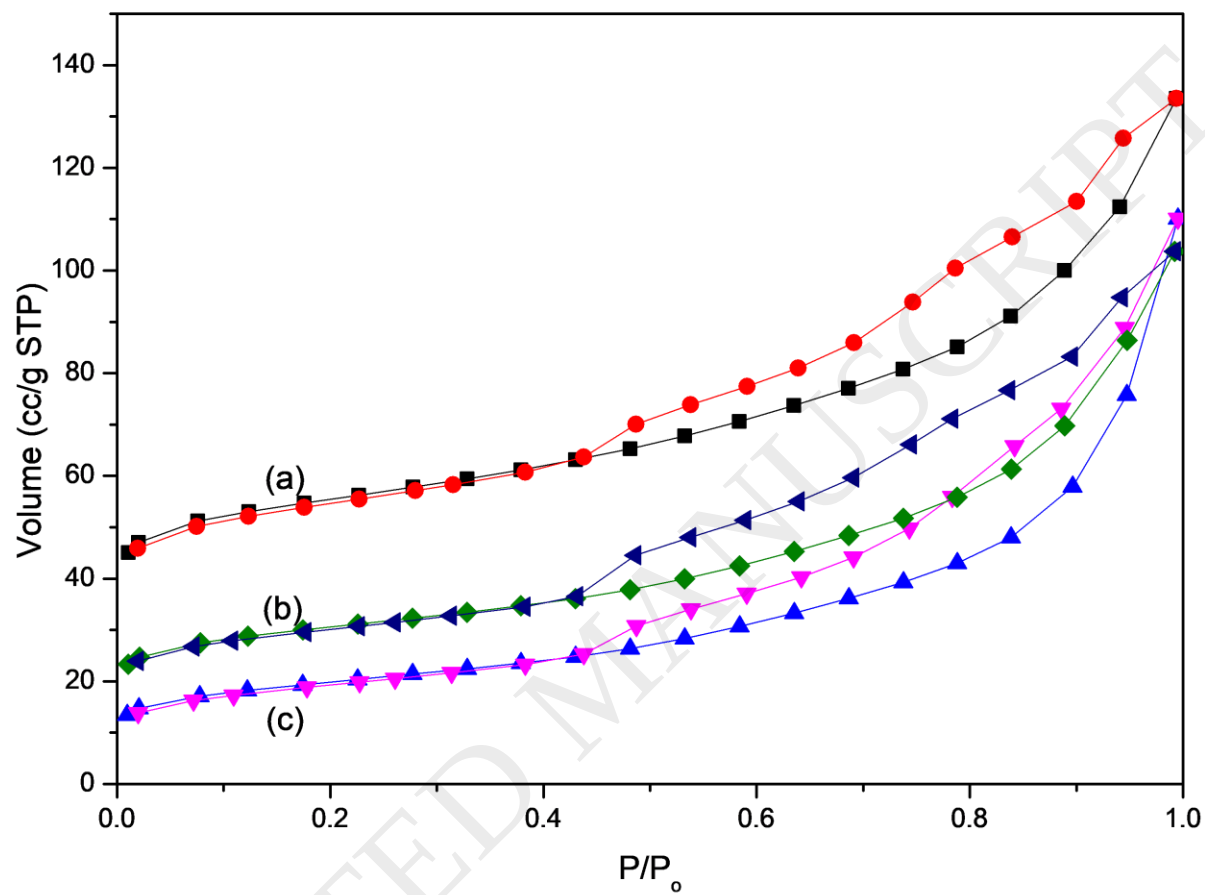


Fig. 11 Correction between micropores surface area and crystallinity of samples with passivator for 3.0, 5.0, 7.5 and 10 wt.% passivator incorporated after drying for 12 h.

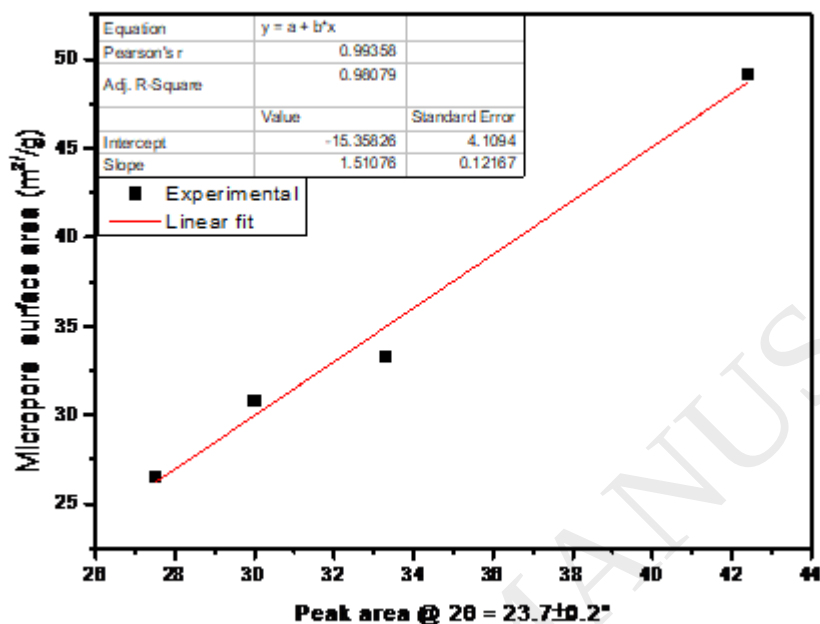


Fig. 13 (a) MAT catalyst activity (b) Effect of catalyst-to-oil ratio on catalyst activity over 10 wt.% Y-Mg-oxide containing catalyst. Insert, effect of temperature on conversion over *7.5 wt.% Y-Mg-oxide containing catalyst. *Passivator added before impregnation of vanadium.

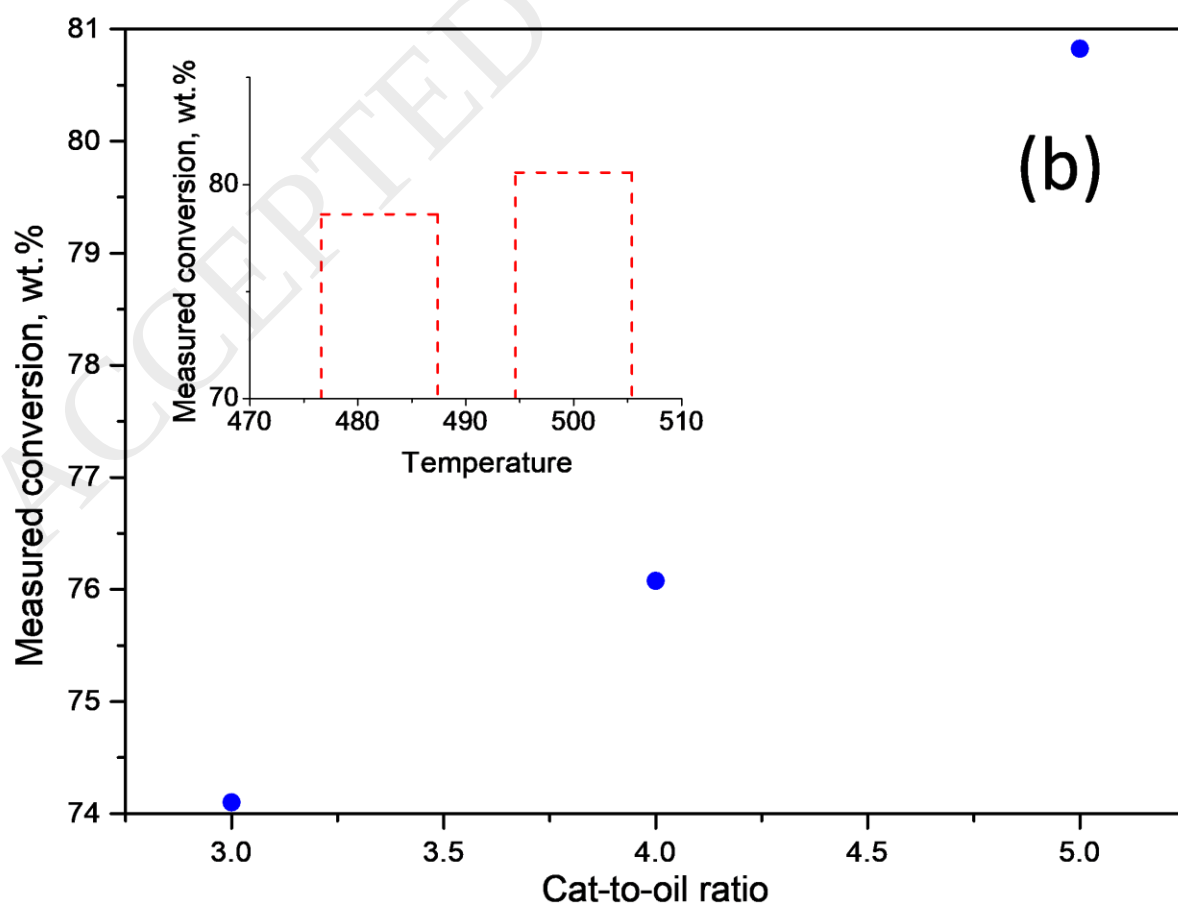
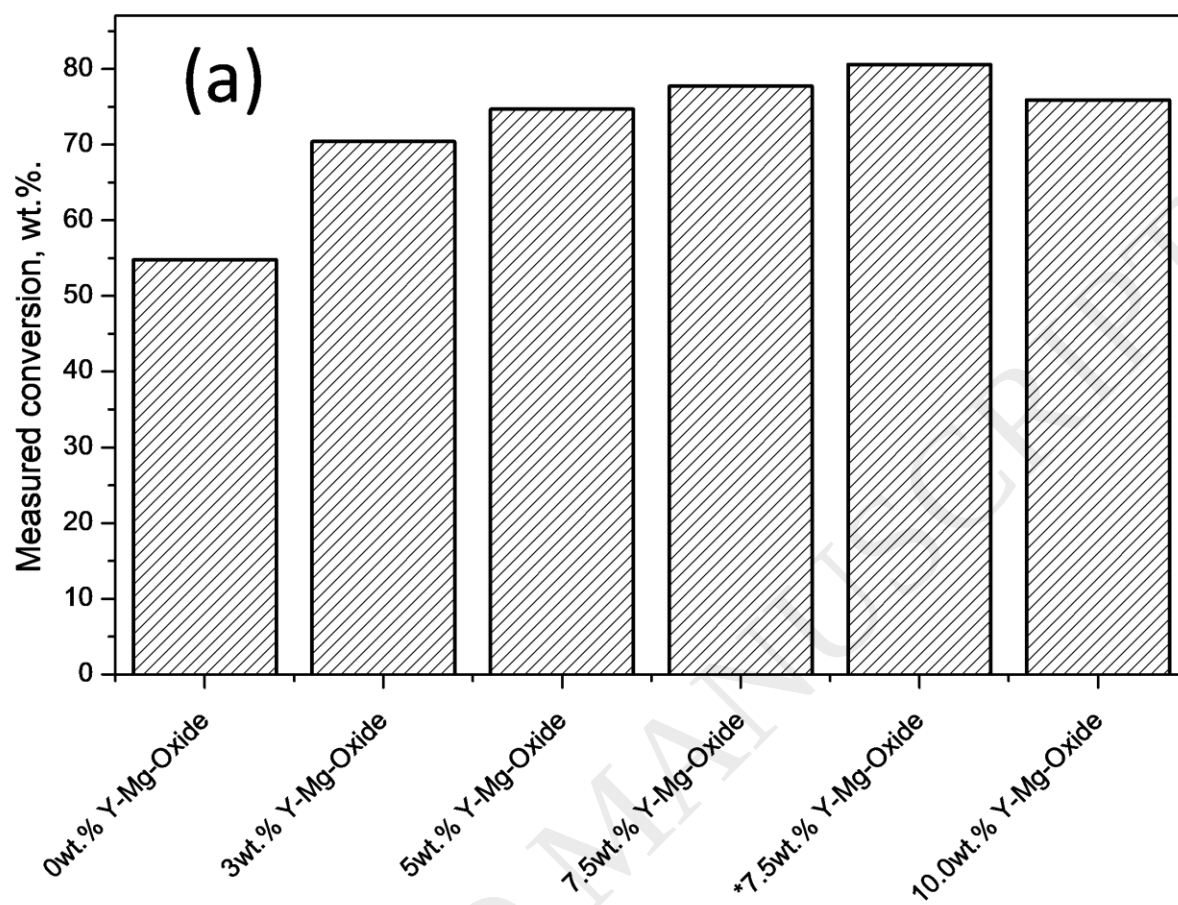


Fig. 14 High resolution pictures of coked catalysts showing passivator particles. (a) Catalyst without passivator, (b) catalyst sample with passivator added before vanadium impregnation, (c) catalyst containing 7.5 wt.% Y-Mg-oxide containing and (d) catalyst containing 10 wt.% Y-Mg-oxide.

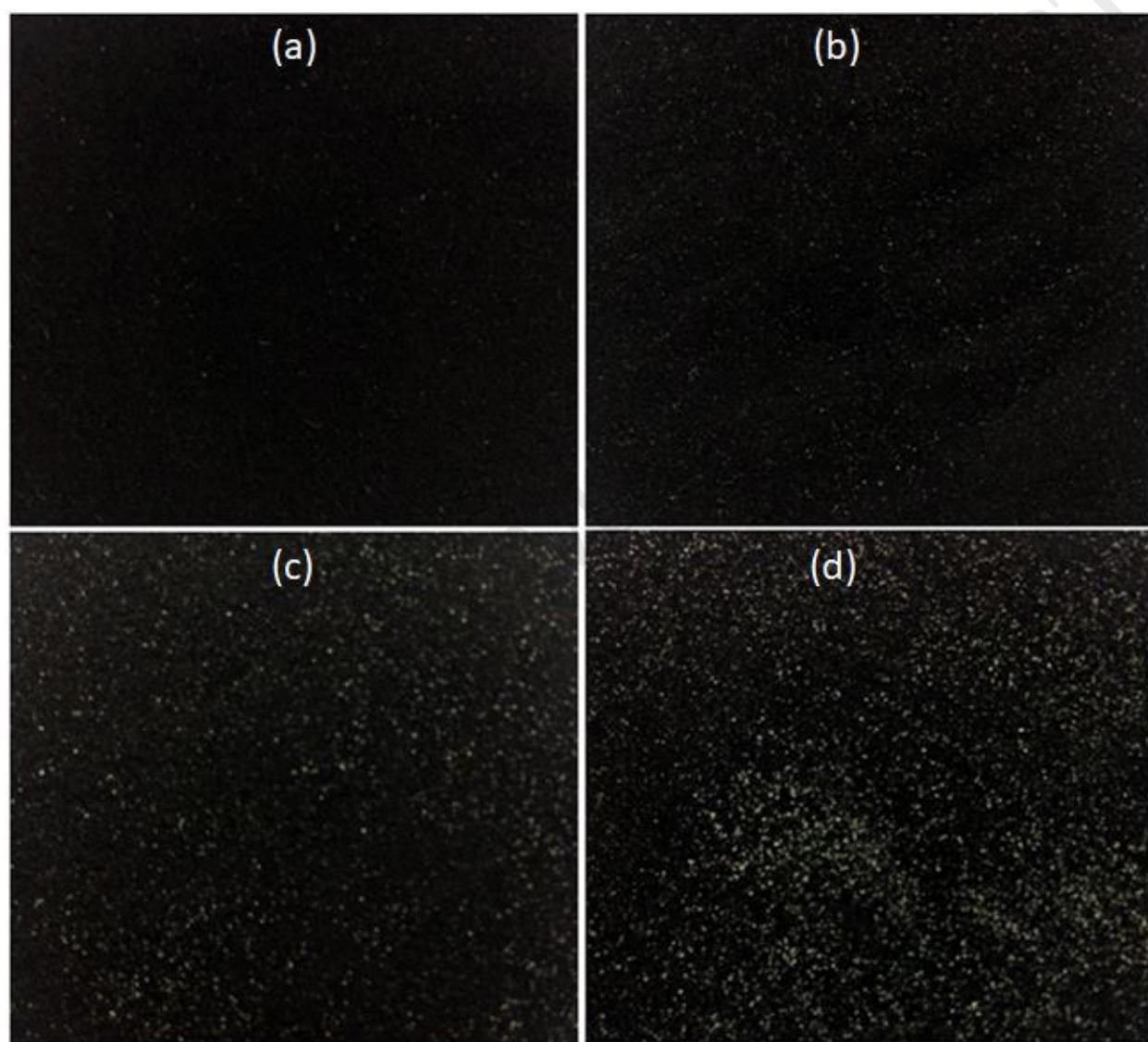
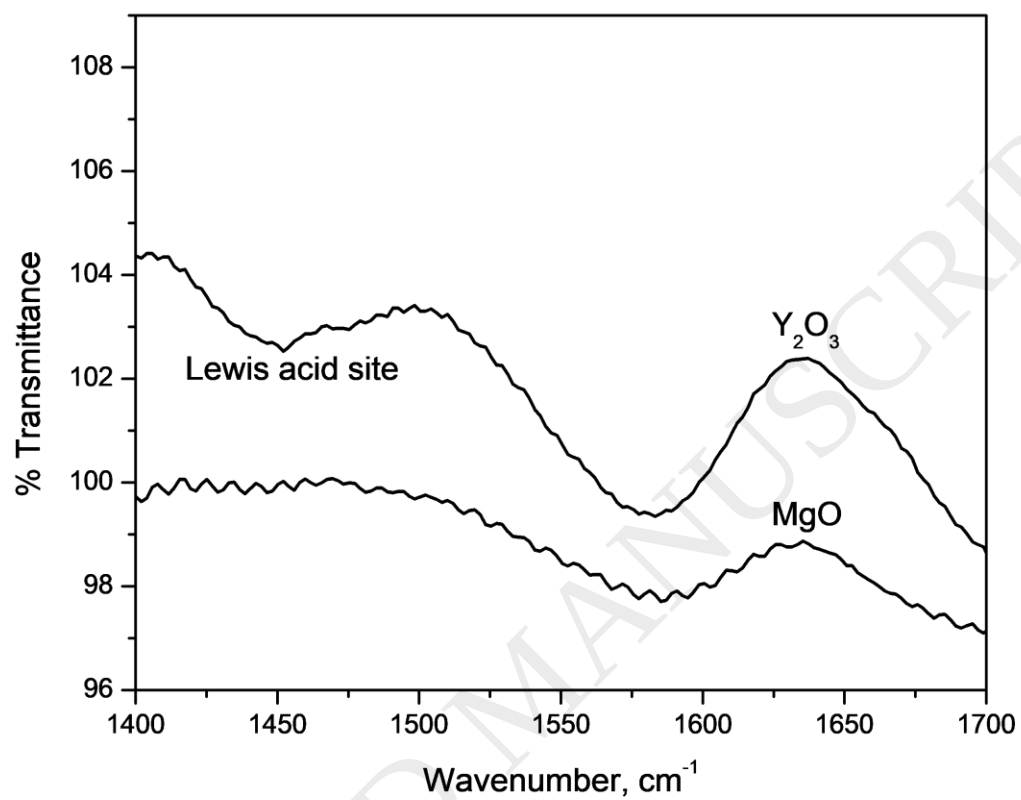


Fig. 15 Transmission IR spectra after pyridine adsorption on metal oxides of the passivator.



Tables

Table 1: Properties of gas oil

Property	Content
Specific gravity @ 25°C (kg/m ³)	921.1
Residual coke (wt.%)	2.32
Asphaltene (wt.%)	0.56
Paraffin (wt.%)	62.07
Aromatics (wt.%)	28.77
Resin (wt.%)	8.6
Metal (ppm)	
Ca	0.84
Fe	2.69
Na	0.52
Ni	3.16
V	4.7

Table 2: Passivator protection degree of vanadium contaminated samples

Passivator (wt.%)	V_{total} (cm ³ /g)	V_{micro} (cm ³ /g)	^a AP	^b AP	D_a (nm)
0	0.14	0.00	–	–	4.5 (23.7)
3.0	0.15	0.01	24.1	16.0	5.3 (23.6)
5.0	0.16	0.02	26.6	20.3	5.3 (22)
7.5	0.18	0.02	29.9	22.8	5.3 (23.7)
10.0	0.18	0.02	39.0	38.7	5.3 (23.9)

^a 7.5	0.16	0.03	42.4	–	4.7 (14.8)
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AP = Activity protection (area of catalyst with passivator – area of catalyst without passivator);

^aPassivator added before impregnation of vanadium; ^acalculated from micropores surface area;

^bcalculated from Eq. 4.

Table 3 EDX elemental scan of 5.0 wt.% Y-Mg-oxide containing catalyst sample single microsphere

Element	Content (wt.%)				
	Point 1	Point 2	Point 3	Point 4	Point 5
O	49.11	46.78	50.32	47.50	51.52
Al	29.62	42.36	36.00	40.46	37.74
Si	10.35	9.92	13.68	12.04	10.74
V	4.51	0.94	–	–	–
Mg	1.10	0.87	–	–	–
Y	5.31	2.32	–	–	–

Table 4: EDX elemental scan of passivator particle in catalyst sample containing 5.0 wt.% Y-Mg-oxide

Element	Content (wt.%)		
	Point 1	Point 2	Point 3
O	25.25	30.99	22.62
Al	10.47	6.99	7.20
Si	13.41	15.62	11.58
V	18.17	8.71	27.89
Mg	31.88	33.18	29.60
Y	0.82	4.50	1.12

Table 5: Normalized product yields

	0wt. %	3.0wt. %	5.0wt. %	7.5wt. %	*7.5wt. %	*7.5wt. %	10.0wt. %	10.0wt. %	10.0wt. %
Catalyst	Y-Mg-	Y-Mg-	%	Y- %	Y- %	Y-Mg-	Y-Mg-	%	Y- %
Sample	oxide	oxide	Mg-oxide	Mg-oxide	oxide (482)	oxide (500)	Mg-oxide(3)	Mg-oxide(4)	Mg-oxide(5)
Product yield (wt.%)									
H ₂	0.23	0.13	0.09	0.06	0.07	0.05	0.05	0.04	0.06
Dry gas	2.01	3.55	2.28	3.44	3.20	1.88	3.78	2.18	1.81
C3 ⁻	0.13	2.93	1.19	0.18	2.83	1.15	2.21	1.64	1.03
C3 ⁼	0.08	2.72	0.35	0.44	2.70	0.70	0.58	0.79	0.67
C4 ⁻	6.44	10.39	8.67	5.45	8.78	5.00	7.12	6.23	6.16
LPG	6.57	13.32	9.86	5.63	11.61	6.14	9.33	8.18	8.51
Gasoline	29.32	34.37	42.48	45.56	45.07	46.80	44.12	45.82	45.18
Diesel	30.08	23.78	22.50	22.79	21.50	23.90	20.84	23.14	22.91
Coke	9.45	8.89	9.92	10.08	9.60	10.92	8.56	9.88	12.01
Total liquid yield	88.74	87.56	87.84	86.48	87.20	87.14	85.64	88.69	86.72
Conversion	47.35	60.13	64.54	64.71	69.48	65.74	65.79	66.06	67.51

$$Conversion = \sum(Dry\ gas: LPG: Gasoline: Coke)$$

$$Total\ Liquid\ yield = \sum(LPG: Gasoline: Diesel: heavy\ oil)$$

(482) and (500) refer to the reactor temperatures and (3, 4 and 5) are catalyst-to-oil ratio.